

Melody: Meta-analysis of Microbiome Association Studies for Discovering Generalizable Microbial Signatures

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Abstract

With the increasing number of microbiome association studies, the field aims to establish standards for data sharing, harmonization, and meta-analysis to accurately identify microbial signatures that are generalizable across cohorts and populations. Standard meta-analysis protocols fall short for microbiome data due to their unique characteristics, particularly their compositional structure, leading to inaccurate and unstable signature selection. To address this issue, we introduce Melody, a framework that generates, harmonizes, and combines summary association statistics across studies to identify microbial signatures in meta-analysis. Comprehensive and realistic simulations demonstrate that Melody substantially outperforms existing approaches in prioritizing true signatures. In applications to five datasets on colorectal cancer and eight datasets on the gut metabolome, we showcase the superior stability, reliability, and predictive performance of Melody-identified signatures.

Significance

Advanced sequencing technologies have accelerated human microbiome research and opened up the opportunity to discover microbial features associated with health outcomes, exposures, and molecular traits. Recent years have witnessed a surge in conducting meta-analyses of microbiome association studies, aiming to identify microbial signatures that are generalizable across cohorts and populations. Standard meta-analysis methods are inadequate for microbiome data that manifest unique characteristics. We have developed a meta-analysis framework to properly account for microbiome data characteristics, enabling the accurate and reliable discovery of generalizable microbial signatures. Extensive simulations and real data applications have demonstrated the advantages of the new method compared to existing approaches.

Introduction

Advanced sequencing technologies have enabled the study of microbial communities that colonize the human body in a culture-independent manner¹. Microbiome association analysis is routinely performed to discover microbial signatures associated with covariates such as health outcomes, exposures, and molecular traits. The signature identified in one study is often hard to generalize to other cohorts and populations because of the small sample size in a single study. A natural way to increase sample size is to perform meta-analyses. Recent years have witnessed a surge in conducting meta-analyses of microbiome association studies²⁻⁵. Unfortunately, standard meta-analysis strategies commonly used in clinical and genomics studies fail to address the unique characteristics of microbiome data, undermining the ability of meta-analysis to identify generalizable microbial signatures.

Microbiome data generated from marker-gene or metagenomics sequencing contain sequencing read counts allocated to various microbial features, such as taxonomic and functional units⁶. The sequencing depth (the total number of reads obtained for a sample) is not correlated with the actual microbial load (the total amount of microorganisms or their DNA volumes) in the sample. Therefore, microbiome data are compositional data that represent the relative abundance (RA) of microbial features. These RA counts are systematically distorted from the underlying absolute abundance (AA) per unit volume^{7,8}. Although techniques have been developed to recover AA data⁹⁻¹¹, the cost is prohibitive and the effectiveness is debatable¹². The RA of all features would shift even if only one microbial feature changes in AA. The small shift may be too subtle to affect the association analysis of a single study but would become more pronounced and impactful as the sample size and data heterogeneity accumulate in the meta-analysis. Several compositionality-aware methods have been developed for microbiome association analysis in a single study¹³⁻¹⁶. These methods can be used for meta-analysis only after properly harmonizing and pooling individual-level data across studies.

In addition to compositionality, other characteristics of microbiome data further compound

the challenges of microbiome association meta-analyses. The sequencing depth strongly influences the RA counts. Methods that fail to consider the variation in sequencing depth across samples can lead to a high number of false positives, particularly when the depths vary significantly between groups of comparison. While rarefaction is commonly used to correct bias introduced by uneven depth, it has been criticized for its tendency to discard valuable data and reduce statistical power. Moreover, for many rare features, insufficient sampling and the absence of features result in an excessive number of zero counts. Some features may be completely missing in a subset of studies in meta-analysis. While the imputation of zero to non-zero values is a widely adopted practice in processing microbiome data, consensus on the best method is lacking even for data from a single study, and it may introduce additional bias to the analysis^{17,18}. Filtering out rare features is another common practice under the belief that rare features are uninformative. However, recent research has noted an alarming phenomenon that analysis with different filtering criteria can result in identifying very different sets of microbial signatures^{19,20}.

Pooling individual-level data and combining summary data are two types of meta-analysis strategies, each of which faces different challenges in microbiome association meta-analyses. Pooled-data meta-analysis requires careful harmonization of individual-level data that usually involves correcting batch effects in microbiome data across studies. Microbiome data can systematically differ among studies involving various age groups, genders, geographic regions, and measurement time points. Batch effects can also occur in sequencing experiments due to differences in reagents, equipment, protocols, and personnel. More importantly, batch effects are often entangled with association effects in microbiome studies^{21–23}. Eliminating batch effects while preserving the signal of interest is difficult²⁴, and there is a potential risk of distorting the signal and adversely influencing downstream analyses^{25,26}.

Summary-data meta-analysis circumvents the need for individual-level data harmonization, as summary statistics are generated for each study separately. This type of meta-analysis is

preferable to pooled-data meta-analysis when there are differences in sample ascertainment and data measurements and availability across studies. If the individual-level data is not shared, the logistical burden and privacy concerns can be significantly reduced, especially when the data contain sensitive information such as genetics or medical history. For these critical reasons, summary-data meta-analysis is widely adopted in many research fields, such as genetic association studies²⁷. However, standard approaches for generating and combining summary statistics are inadequate for microbiome data due to their unique characteristics mentioned above.

We introduce a summary-data meta-analysis framework named Melody (Meta-analysis for selecting microbial drivers in compositionality). Melody generates accurate and robust summary association statistics between the RA of microbial features and the covariate of interest by respecting the microbiome data characteristics. The procedure does not require microbiome data normalization, rarefaction, zero imputation, or batch effect correction across studies. By harmonizing and combining the summary statistics across studies, Melody identifies microbial signatures with their AA consistently associated with the covariate in all studies. Melody allows study-specific confounder adjustments, accommodates correlated samples, and is computationally efficient.

The recovery of the AA association in Melody is achieved by the assumption that the true signatures are “drivers”, defined as the most parsimonious set of microbial features with their AA associations compatible with the observed RA associations. Fig. S1 illustrates that the underlying AA association profile compatible with the same RA association profile (referred to as “putative AA association”) are not unique, leading to fluidity in the derived signatures. The upper panel of the figure shows the RA of microbial features in two groups and the RA association profile when contrasting the RA of the two groups; the lower panel shows three example scenarios of putative AA association profile. In this illustration, the driver signatures are derived from the first scenario because it produces the most

parsimonious association signals among all possible versions of the putative AA association profile. We will demonstrate that focusing on detecting drivers is essential to harmonize RA summary statistics across studies and identify stable and generalizable microbial signatures in meta-analysis.

Results

Melody overview

Melody is described in *Materials and Methods*, its technical details are provided in the *Supplementary Note*, and its schematic overview is shown in Fig. 1. For each study, we link the RA count data with the covariate of interest using a quasi-multinomial regression model and obtain summary statistics consisting of the estimates of the RA association coefficients and their variances. The RA association coefficients are estimated based on the scaled score statistics, which are numerically stable, statistically accurate, and computationally efficient^{28,29}. The quasi-multinomial framework allows the arbitrary distribution of the underlying microbial compositions and guards against inaccurate inference from distribution misspecification. The model easily accommodates confounder adjustments and naturally handles correlated samples (e.g., from longitudinal studies). The RA association coefficients in the model are formulated at the log-ratio scale, with a specified feature as the reference. The RA association coefficients are expressed as the shifted AA association coefficients by a constant (*Materials and Methods*).

Next, we combine the RA summary statistics across studies to estimate the sparse meta AA association coefficients. This meta-analysis is framed as a best subset selection problem^{30,31}, where we employ a cardinality constraint with the subset size hyperparameter s to encourage the sparsity of meta AA association coefficients. To recover meta coefficients at the AA level using summary statistics at the RA level, we introduce a hyperparameter δ_ℓ in each study ℓ to shift the RA association coefficient estimates, producing different versions of putative AA

association coefficients by varying δ_ℓ . These study-specific AA coefficients are then combined to estimate meta AA coefficients. The hyperparameters s and δ_ℓ 's are jointly tuned using the Bayesian information criterion (BIC) to balance the trade-off between the model fit and sparsity. As illustrated in Fig. 1, the optimal δ_ℓ values shift the RA coefficients by an appropriate amount to identify a version of putative AA association coefficients that are sparse and consistent across studies. The Melody-identified driver signatures are microbial features with non-zero estimates of meta AA association coefficients under the optimal hyperparameters.

It is important to note that while we need to select a reference feature for generating summary statistics, the signature selection is not sensitive to the choice of the reference. We can even choose different references for different studies (see the sensitivity analysis in the real data application). This is because the reference effects across studies are offset by δ_ℓ 's in the process of estimating the sparse meta AA association coefficients.

Simulations

In the simulations, we compared Melody with the MMUPHin summary-data meta-analysis method and several pooled-data meta-analysis approaches for selecting microbial signatures with differential abundance across two groups of interest. MMUPHin⁵ is an R package that includes a batch-effect correction algorithm and a summary-data meta-analysis method. The meta-analysis first generates summary statistics using MaAsLin2³² within individual studies and then combines these summary statistics using standard meta-analysis methods³³ for association testing. In the pooled-data meta-analysis, the pooled data were analyzed using representative microbiome association tests ALDEx2³⁴, ANCOM-BC2¹⁶, and the blocked Wilcoxon test (BW)³, as well as a penalized regression method CLR-LASSO³⁵. In ALDEx2 and ANCOM-BC2, we adjusted for studies as confounders. In BW, we rarefied count data within each study and performed the blocked Wilcoxon rank-sum test with studies

specified as blocks. In CLR-LASSO, we used the centered log-ratio (CLR) transformed microbiome data as predictors in the LASSO regression³⁶. Melody was applied to the original data, and the compared methods were applied to both the original data and study-effect corrected data using the MMUPHin batch-effect correction algorithm. We compared the performance using the area under the precision-recall curve (AUPRC) for signature selection. Specifically, the precision-recall curves of the test-based methods (MMUPHin, BW, ALDEx2, ANCOM-BC2) were constructed by changing the p -value cutoff, and those of the penalized methods (Melody and CLR-LASSO) were constructed by varying the hyperparameter that controls the signature sparsity level.

We simulated microbiome data for five studies that resemble the structure of five real microbiome datasets³(Fig. S2). The detailed simulation strategy is described in *Materials and Methods*. In brief, we consider a wide range of settings by varying the study sample size, the number of signatures, the degree of sign imbalance among signatures, and the degree of sequencing depth unevenness between the two groups of interest. Melody produces the highest AUPRC values in all settings and is robust with varying degrees of sign imbalance and sequencing depth unevenness (Fig. 2). The AUPRC of ANCOM-BC2 reduces substantially as the degrees of the two factors increase. The AUPRC of MMUPHin, CLR-LASSO, BW, and ALDEx2 reduce significantly as the degree of sign imbalance increases. When using the study-effect corrected data, the AUPRC of MMUPHin slightly increases, while the pooled-data meta-analysis methods show minimal change. We performed another set of simulations for the meta-analysis of studies with correlated samples and compared Melody with MMUPHin and ANCOM-BC2, which can handle correlated samples. The patterns of the results remain the same (Fig. S3).

Meta-analysis of microbiome association studies for colorectal cancer

We conducted a meta-analysis of five metagenomics studies^{3,37–40} for colorectal cancer (CRC) to identify microbial signatures associated with CRC. The data contain 849 gut bacterial

species measured on 574 fecal samples across the five studies (Table S1). Microbial communities display a pattern of cross-study heterogeneity (Fig. S2). The sequencing depth between CRC cases and controls is significantly different in some studies (Fig. S4). Our main analysis focused on 401 microbial species with at least 20% prevalence across all samples in the five studies. We applied the methods that were compared in the simulations to analyze the data. For the test-based methods, the signatures were selected using the Benjamini-Hochberg (BH) procedure⁴¹ at the false discovery rate (FDR) level of 0.05. For CLR-LASSO, the signatures were selected based on the best LASSO model determined by five-fold cross-validation.

The meta-analysis results of all methods are presented in Table S2. Melody identified 51 signatures for CRC (Fig. 3a), with the top two signatures *Fusobacterium nucleatum* and *Parvimonas micra* being the well-known species promoting CRC^{42,43}. Using original data, MMUPHin, CLR-LASSO, BW, ALDEx2, and ANCOM-BC2 selected 84, 94, 110, 44, and 72 signatures, respectively. Signatures identified on the study-effect corrected data are similar to those identified on the original data. The top-selected species and their rankings differ significantly across methods (Fig. 3b, Figs. S5 and S6).

In Melody analysis, we used species *Coproccoccus catus* as the reference when generating summary statistics for individual studies. We also chose other references and evaluated the sensitivity of the Melody result. Moreover, we used different references for different studies and repeated the analysis. The Melody signatures are highly consistent across these reference configurations (Fig. 3c). Although the RA association coefficient summary statistics show noticeable differences across various choices of reference, the Melody-recovered AA association coefficient estimates are almost invariant (Fig. S7). The AA coefficients also show better harmonization with consistent association directions across studies than the RA coefficients.

We evaluated the stability of signature selection from three perspectives. (1) Is the selection stable under different subsets of the samples? We selected signatures on leave-one-study-out

(LOSO) data (i.e., the data with one study left out) and evaluated the overlap of the signatures with those selected on data of all studies. (2) Is the selection stable after applying different prevalence filters? Our main analysis was on $K = 401$ species after applying the 20% prevalence filter. We also identified signatures based on all $K = 849$ species, kept the subset of the signatures within the $K = 401$ species list, and evaluated the overlap of the subset with those selected in the main analysis. (3) Is the selection stable if analyzing a subcomposition? We identified signatures based on a subcomposition with $K = 267$ species under order *Clostridiales*. We evaluated the overlap of the identified signatures with the subset of main-analysis-identified signatures within the $K = 267$ species list. We bootstrap the complete data of five studies 100 times, repeat these stability evaluations on the bootstrap samples, and report the Jaccard index to measure set overlap (Fig. 4). When the subset of samples included in the meta-analysis is changed, Melody and BW exhibit higher stability compared to other methods. When the set of microbial features included in the meta-analysis is changed, the stability of Melody is much higher than that of BW.

In addition, we evaluated the out-of-sample prediction power of the top-selected species using the LOSO strategy. We held out one study for prediction evaluation and randomly split samples from the remaining four studies into a feature selection set and a prediction model training set at a sample size ratio of 3:1. We performed the meta-analysis on the feature selection set to pick the top N CRC-associated species. We then used the sample proportions of these species in the training set to train a random forest model for CRC prediction. Finally, we calculated the area under the receiver operating characteristic (AUROC) of the best random forest model on the hold-out study. We repeated the procedure 100 times with different random splits of the samples and reported the average AUROC (Fig. 5). The Melody-identified top species have substantially higher AUROC than those identified by the test-based methods in CRC1, 2, and 5, and are comparable to the highest AUROC in other studies. The top species identified by CLR-LASSO performed well in CRC1 but much worse in CRC2, 4, and 5. Overall, the group of species prioritized by Melody has

the highest average AUROC in CRC prediction, whereas the species prioritized by other methods yielded similar results (Combined panel in Fig. 5).

Meta-analysis of microbiome-metabolome association studies

We conducted another meta-analysis of eight microbiome-metabolome association studies^{44–51} to identify microbial signatures associated with human gut metabolites. Most of these studies are case-control studies with one group of individuals with a specific medical condition and the other group of healthy control individuals (Table S3). The raw data contain 12,195 gut bacterial genera and 1,151 HMDB-annotated⁵² metabolites measured on 2,127 fecal samples across the eight studies. We focused on metabolites and genera with at least 10% prevalence within individual studies. Most metabolites are available in only a subset of studies (Fig. S8). We considered metabolites that are present in at least two studies. The final datasets for analysis contain 450 metabolites and 101 genera, resulting in 43,722 metabolite-genus pairs. There is notable heterogeneity among studies in study designs, microbiome sequencing methods, and metabolite measurements (Table S3), rendering pooled-data meta-analysis unsuitable. Therefore, we performed Melody and MMUPHin summary-data meta-analyses on the original data. In both methods, we analyzed each metabolite variable as a covariate of interest while adjusting the disease status as a confounder when generating summary statistics for individual studies. For MMUPHin, the signatures for each metabolite were selected using the BH procedure at the FDR level of 0.05.

Melody completed the analysis within 20 minutes, while MMUPHin took 25 minutes on a laptop (Apple M1 Pro 3.2GHz processor). Melody identified 5,847 metabolite-genus pairs, among which 2,499 have positive and 3,348 have negative associations (Table S4). MMUPHin identified 5,921 pairs, among which 2,065 have positive and 3,856 have negative associations (Table S5). Fig. 6 displays the meta association coefficient estimates of the two methods for the same set of top metabolites, which are associated with a large number of genera. Melody prioritizes many more genera than MMUPHin that have disproportionately

positive associations with the top metabolites. These genera are associated with much fewer metabolites in MMUPHin results (Fig. S9.b). Several of these genera, such as *Anaerostipes*, *Blautia*, and *Roseburia*, are known to play essential roles in various metabolic processes such as fermentation of dietary fibers and various types of carbohydrates^{53,54}. These genera have also been linked with multiple human complex diseases such as metabolic syndrome, inflammatory bowel disease, and obesity⁵⁵⁻⁵⁷. Besides the direct engagement of the genus in metabolite production, consumption, or degradation, correlations between metabolites and genera can also result from indirect associations, such as interactions between gut bacteria or co-abundant metabolites.

We further examined metabolites for which the results of the two methods were strikingly different. Fig. S10 focuses on such a metabolite for which Melody identified 13 positive-associated genera and 10 negative-associated genera, whereas MMUPHin identified 7 positive-associated genera and 21 negative-associated genera. For each method, we removed 5 genera with the highest positive meta association identified by the method and reran the meta-analysis. The Melody signature selection is stable after the genus removal. MMUPHin identified many new positive-associated genera, and the set of negative-associated genera also changed substantially. This is a global phenomenon, not limited to a few metabolites. We performed the same evaluation for all 106 metabolites with at least 5 positive-associated genera detected by Melody and MMUPHin. The overlap of the signatures before and after genus removal is much higher in Melody than in MMUPHin for most of these metabolites (Fig. S11). The stability of the Melody results shown in this experiment is mainly attributed to its ability to capture the fluid association signals in the compositional data. This evaluation underscores the importance of addressing compositionality to ensure the stability and reliability of results in microbiome association meta-analyses.

Discussion

We have developed Melody to address major challenges in the meta-analysis of microbiome

association studies. Melody’s key innovations are (a) generating accurate and robust summary statistics by accounting for the unique characteristics of microbiome data; (b) recovering meta association coefficients at the absolute abundance level and identifying driver signatures consistent across studies. Melody has a unique combination of features that make it desirable for practical use: (1) no need to share individual-level data or correct study effects; (2) no need to perform data normalization, rarefaction, or zero imputation; (3) no need to assume that the same set of microbial features is measured and analyzed across studies; (4) allows for different designs across studies (e.g., some studies are cross-sectional, and some are longitudinal); (5) easily adjusts for confounders and allows different sets of confounders available across studies; (6) fast computation for large-scale association scans.

As demonstrated in the real data application, the set of Melody-identified signatures is not sensitive to the prevalence filter. However, like any method for high-dimensional feature selection, including lots of noise features could adversely impact the selection accuracy. Therefore, we still suggest applying a prevalence filter before data analysis if investigators are not interested in detecting very rare features. The filter can be applied to pooled samples of all studies (as in the meta-analysis for CRC) or samples within each study in case a feature may be more prevalent in some studies and worth keeping (as in the meta-analysis for metabolome). Melody naturally accommodates different sets of features across studies in meta-analysis.

We utilized the MMUPHin batch-effect correction algorithm to correct for study effects in our numerical studies and found that the accuracy of the signature selection based on the study-effect corrected data is similar to that based on the original data. We also investigated another algorithm⁵⁸ in simulations (Fig. S12) and found that the accuracy of signature selection using the study-effect corrected data was dramatically reduced compared to using the original data. This suggests that batch effect correction has the potential to distort association signals and adversely affect downstream association analysis.

319 This article focuses on identifying microbial signatures that are shared across studies, as these
320 are more likely to be generalizable to broader populations. It can also be of interest to identify
321 microbial signatures with heterogeneous associations across populations (e.g., microbial
322 features associated with the covariate in females but not in males)⁵⁹. Melody can be extended
323 to detect such heterogeneous associations by adapting random-effects meta-analysis and
324 meta-regression techniques⁶⁰. With the ever-increasing number of microbiome datasets, we
325 anticipate that Melody will facilitate the accumulation of sufficiently large sample sizes to
326 accelerate the reliable discovery of generalizable microbiome signatures.

Materials and Methods

Melody framework. *Generating summary statistics for individual studies.* The first step of Melody is to obtain summary association statistics between microbial composition and the covariate of interest in each participating study. Suppose we have n independent microbiome samples with K features in a given study. For sample i , let X_i denote the covariate, N_i denote the sequencing depth of the microbiome data, Y_{ik} denote the number of sequencing reads assigned to feature k (i.e., RA count data), and W_{ik} denote the unobserved AA of feature k . It is natural to model the mean of AA as

$$E(W_{ik}) = \exp(\beta_{0k} + \beta_k X_i), \quad k = 1, \dots, K,$$

where β_{0k} is the intercept and β_k represents the AA association coefficient for feature k . The relation between the RA mean and the AA mean can be expressed as

$$E(Y_{ik}) = c_i b_k E(W_{ik}) = c_i b_k \exp(\beta_{0k} + \beta_k X_i), \quad k = 1, \dots, K, \quad (1)$$

where c_i is the sample-level multiplicative factor that absorbs quantities such as the sequencing depth and the microbial load in a unit volume for a sample, and b_k is the feature-level multiplicative factor that absorbs the measurement biases arising from the different efficiencies (e.g., DNA extraction efficiency, primer binding, and amplification efficiency) with which various features are measured¹².

The AA coefficients are not identifiable in the model (1). Therefore, we choose a feature R as the reference and construct the proportion mean parameter as

$$p_{ik} = \frac{E(Y_{ik})}{\sum_{j=1}^K E(Y_{ij})} = \frac{\exp(\beta_{0k}^{(R)} + \beta_k^{(R)} X_i)}{1 + \sum_{j \in \setminus R} \exp(\beta_{0j}^{(R)} + \beta_j^{(R)} X_i)}, \quad (2)$$

$$\beta_{0k}^{(R)} = \log(b_k) + \beta_{0k} - \log(b_R) - \beta_{0R} \text{ and } \beta_k^{(R)} = \beta_k - \beta_R, \quad k \in \setminus R, \quad (3)$$

where $\setminus R$ is the index set of non-reference features, and $\beta_{0k}^{(R)}$'s and $\beta_k^{(R)}$'s are RA coefficients. Note that the RA association coefficients of the non-reference features $\beta_k^{(R)}$'s are simply shifted AA association coefficients β_k 's by a constant β_R .

352 The model can be extended to incorporate confounding variables. Let $\mathbf{X}_i$ denote the design
 353 matrix that includes the intercept and all the explanatory variables and $\boldsymbol{\theta}^{(R)} = (\boldsymbol{\beta}^{(R)}, \boldsymbol{\gamma}^{(R)})$
 354 be the collection of all RA coefficients, in which $\boldsymbol{\beta}^{(R)}$ contain the RA association coefficients
 355 of interest and $\boldsymbol{\gamma}^{(R)}$ contain the nuisance coefficients.

356 We propose a quasi-multinomial model to estimate the RA coefficients. Specifically, given
 357 the true proportions of various features $(P_{i1}, \dots, P_{iK})$, the RA counts $(Y_{i1}, \dots, Y_{iK})$ naturally
 358 arise from multinomial sampling with parameters $(P_{i1}, \dots, P_{iK})$ and size N_i . The multinomial
 359 distribution models microbial features competing for the sequencing reads bounded by the
 360 sequencing depth N_i (i.e., allocating more reads to one feature means that fewer reads
 361 are available to assign to other features). The multinomial layer is particularly important
 362 for properly accounting for the effect of uneven sequencing depth on the variability of the
 363 RA counts given the true proportions. To address the over-dispersion, zero-inflation, and
 364 complex structure of the microbiome data, an additional layer of randomness is needed for
 365 $(P_{i1}, \dots, P_{iK})$. We do not specify a particular distribution for this layer but only assume the
 366 proportion mean model $E(P_{ik}) = p_{ik}$ specified in (2). To estimate the RA coefficients, we
 367 propose the following quasi-score function

$$368 \quad \mathbf{S}(\boldsymbol{\theta}^{(R)}) = \sum_{i=1}^n (\mathbf{Y}_i - N_i \mathbf{p}_i) \otimes \mathbf{X}_i,$$

369 where $\otimes$ represents the Kronecker product, $\mathbf{Y}_i = \{Y_{ik}\}_{k \in \mathcal{R}}$, and $\mathbf{p}_i = \{p_{ik}\}_{k \in \mathcal{R}}$. The
 370 parameters can be consistently estimated by solving the estimating equation $\mathbf{S}(\boldsymbol{\theta}^{(R)}) = \mathbf{0}$.
 371 This is equivalent to finding the maximum likelihood estimator from multinomial logistic
 372 regression because the quasi-score function has the same form as the score function of
 373 multinomial regression. To prevent the potential estimation bias due to the small sample
 374 size and the presence of rare features, we adopted the mean bias reduction procedure in
 375 estimating the parameters of the multinomial regression⁶¹.

376 Let $\mathbf{S}_\beta$ and $\mathbf{S}_\gamma$ denote the components in $\mathbf{S}$ corresponding to $\boldsymbol{\beta}^{(R)}$ and $\boldsymbol{\gamma}^{(R)}$, respectively.
 377 Score statistics have been shown to be more accurate and numerically stable than Wald

378 statistics in genetic association studies²⁸. In light of this, we propose estimating $\beta^{(R)}$ based
 379 on the scaled score statistics

$$380 \quad \hat{\beta}^{(R)} = \mathcal{I}_{\beta}^{-1}(\tilde{\theta}^{(R)}) \mathcal{S}_{\beta}(\tilde{\theta}^{(R)}),$$

381 where $\tilde{\theta}^{(R)} = (\mathbf{0}, \tilde{\gamma}^{(R)})$ and $\tilde{\gamma}^{(R)}$ is the estimate of $\gamma^{(R)}$ under the reduced model without the
 382 covariate of interest in $\mathbf{X}_i$, $\mathcal{I}_{\beta} = \mathcal{I}_{\beta\beta} - \mathcal{I}_{\beta\gamma} \mathcal{I}_{\gamma\gamma}^{-1} \mathcal{I}_{\gamma\beta}$, and $\begin{bmatrix} \mathcal{I}_{\beta\beta} & \mathcal{I}_{\beta\gamma} \\ \mathcal{I}_{\gamma\beta} & \mathcal{I}_{\gamma\gamma} \end{bmatrix}$ is the partition of the
 383 quasi-information matrix $-\frac{\partial \mathcal{S}(\theta^{(R)})}{\partial \theta^{(R)}}$ for $\beta^{(R)}$ and $\gamma^{(R)}$.

384 The sandwich estimator of the covariance of $\hat{\beta}^{(R)}$ can be written as

$$385 \quad \mathbf{V}^{(R)} = \mathcal{I}_{\beta}^{-1}(\tilde{\theta}^{(R)}) \left(\sum_{i=1}^n \mathbf{U}_i(\tilde{\theta}^{(R)}) \mathbf{U}_i^T(\tilde{\theta}^{(R)}) \right) \mathcal{I}_{\beta}^{-1}(\tilde{\theta}^{(R)}),$$

386 where $\mathbf{U}_i = \mathcal{S}_{\beta,i} - \mathcal{I}_{\beta\gamma} \mathcal{I}_{\gamma\gamma}^{-1} \mathcal{S}_{\gamma,i}$, $\mathcal{S}_{\beta,i}$ and $\mathcal{S}_{\gamma,i}$ are the i th summand of $\mathcal{S}_{\beta}$ and $\mathcal{S}_{\gamma}$ respectively.

387 The correlations cannot be accurately estimated when the number of features is close to
 388 or larger than the sample size. Therefore, we use the diagonal matrix (with variances
 389 on the diagonal elements) extracted from $\mathbf{V}^{(R)}$ in the meta-analysis. This procedure of
 390 generating summary statistics can be modified for studies with correlated samples (e.g.,
 391 from longitudinal studies), and the details are provided in the *Supplementary Note*.

392 *Harmonizing and combining summary statistics across studies to identify signatures.* Suppose
 393 we want to meta-analyze L independent microbiome association studies and identify microbial
 394 signatures shared by the studies. Let $\beta_{\bullet\ell}$ and $\beta_{\bullet\ell}^{(R)}$ respectively denote the AA and RA
 395 association coefficients for study ℓ . The length of these coefficients could differ among
 396 studies because a feature may not be observed in all studies. Let $\mathcal{T}_{\ell}$ be the index set of
 397 non-reference features observed in study ℓ . For each study ℓ , we obtain RA association
 398 coefficient estimates denoted by $\hat{\beta}_{\bullet\ell}^{(R)} = \{\hat{\beta}_{k\ell}^{(R)}\}_{k \in \mathcal{T}_{\ell}}$ and the diagonal matrix $\mathbf{V}_{\ell}^{(R)}$ in which
 399 the diagonal elements are the variance of $\hat{\beta}_{\bullet\ell}^{(R)}$. In the meta-analysis, we are interested in
 400 estimating the meta AA association coefficients that are common across all study-specific
 401 AA coefficients $\beta_{\bullet\ell}$ ($\ell = 1, \dots, L$). Let $\mathcal{T}_0$ denote the union set of all $\mathcal{T}_{\ell}$'s, K_0 denote the size
 402 of $\mathcal{T}_0$, and $\mu = \{\mu_k\}_{k \in \mathcal{T}_0}$ be the collection of the meta AA association coefficients for all the

non-reference features appear in at least one study. We propose to estimate $\boldsymbol{\mu}$ by minimizing the objective function

$$\mathcal{L}(\boldsymbol{\mu}) = n_0^{-1} \sum_{\ell=1}^L (\hat{\boldsymbol{\beta}}_{\bullet\ell}^{(R)} + \delta_\ell \mathbf{1}_\ell - \boldsymbol{\mu}_{\mathcal{T}_\ell})^T \mathbf{V}_\ell^{(R)-1} (\hat{\boldsymbol{\beta}}_{\bullet\ell}^{(R)} + \delta_\ell \mathbf{1}_\ell - \boldsymbol{\mu}_{\mathcal{T}_\ell}), \text{ subject to } \|\boldsymbol{\mu}\|_0 \leq s, \quad (4)$$

where n_0 is the total sample size across studies, $\mathbf{1}_\ell$ is a K_ℓ -vector of ones, $\boldsymbol{\mu}_{\mathcal{T}_\ell}$ is the subset of $\boldsymbol{\mu}$ for features in $\mathcal{T}_\ell$, and $\|\boldsymbol{\mu}\|_0 = \sum_{k=1}^{K_0} I(\mu_k \neq 0)$ is the ℓ_0 norm of $\boldsymbol{\mu}$. The form of $\mathcal{L}(\boldsymbol{\mu})$ is motivated by the inverse-variance estimator commonly used in meta-analyses⁶², and the cardinality constraint with the sparsity level s is formulated as in the classical best subset selection problem^{30,31}. By noting the relationship between AA and RA association coefficients $\beta_{k\ell}^{(R)} = \beta_{k\ell} - \beta_{R\ell}$ ($k \in \mathcal{T}_\ell$), we introduce a hyperparameter δ_ℓ in each study ℓ to offset $\beta_{R\ell}$ in $\beta_{k\ell}^{(R)}$. By varying the value of δ_ℓ , one essentially traverses possible versions of putative AA association coefficients in study ℓ . Given $\delta_1, \dots, \delta_L$ and s , we solve the best subset selection problem (4) using the polynomial algorithm⁶³. The optimal values of the hyperparameters are determined by minimizing BIC defined as

$$\text{BIC}(s, \delta_1, \dots, \delta_L) = n_0^{-1} \sum_{\ell=1}^L (\hat{\boldsymbol{\beta}}_{\bullet\ell}^{(R)} + \delta_\ell \mathbf{1}_\ell - \tilde{\boldsymbol{\mu}}_{\mathcal{T}_\ell})^T \mathbf{V}_\ell^{(R)-1} (\hat{\boldsymbol{\beta}}_{\bullet\ell}^{(R)} + \delta_\ell \mathbf{1}_\ell - \tilde{\boldsymbol{\mu}}_{\mathcal{T}_\ell}) + \|\tilde{\boldsymbol{\mu}}\|_0 \frac{\log n_0}{n_0}, \quad (5)$$

where $\tilde{\boldsymbol{\mu}}$ is the estimate of $\boldsymbol{\mu}$ under a specification of hyperparameters. Details on solving the best subset selection problem and hyperparameter tuning algorithms are provided in the *Supplementary Note*. The $\tilde{\boldsymbol{\mu}}$ under the optimal hyperparameter values is the estimate of the most parsimonious version of the putative AA associations consistent across studies. The features with non-zero values in $\tilde{\boldsymbol{\mu}}$ are the driver signatures detected by Melody.

Simulation strategy. We simulated microbiome data by mimicking the species composition patterns of the five metagenomics studies for CRC³. For each study, we first fit the generalized Dirichlet multinomial distribution⁶⁴ to the 401 species with a prevalence of at least 20% in the real data and obtained the maximum likelihood parameter estimates. Using the five sets of estimated parameters, we randomly generated independent samples of proportions for five studies from the generalized Dirichlet distribution. These samples served as the true

428 proportions of the microbial features. The synthetic data closely resemble the structure of
 429 the real microbiome data (Fig. S2). We considered two settings for the sample sizes of the
 430 five studies: $n = 100, 120, 140, 160, 180$ for relatively large studies, and $n = 20, 30, 40, 50, 60$
 431 for small studies. We randomly assigned the samples into 50% cases and 50% controls.
 432 We varied the number of signatures from 20 to 80. Half of the signatures were randomly
 433 selected from the set of relatively abundant features with an average proportion of at least
 434 10^{-3} in the original real data, while the other half were randomly selected from the remaining
 435 features. We let $pos\%$ of the signatures have positive effects and $1 - pos\%$ have negative
 436 effects. We considered three settings of $pos\%$: 60%, 80%, and 100%, which reflect various
 437 degrees of sign imbalance among signatures. For each positive-effect signature, we increased
 438 their proportions in cases; for each negative-effect signature, we increased their proportions
 439 in controls. The increased fold for each signature was randomly sampled from $\text{Uniform}(0,$
 440 $\Delta)$, where $\Delta = 2$ under the large study setting and $\Delta = 5$ under the small study setting.
 441 Next, we normalized the proportions of all features in each sample to ensure that the total
 442 sum of all proportions is one. Notably, the proportions of all features are shifted after this
 443 step, but the features initially changed are considered the driver signatures with different
 444 underlying absolute abundances between the case and control groups. We then used the
 445 normalized proportions as multinomial parameters to sample count data. The sequencing
 446 depth for samples in each study was randomly drawn from the pool of sequencing depth
 447 in the corresponding CRC study. Within each study, we increased the sequencing depth
 448 of all samples in either case or control group by a factor u to reflect the potential uneven
 449 sequencing depth between the two groups of interest. We considered three settings of u : 0,
 450 0.5, and 1. We simulated 100 datasets for each scenario and reported the average AUPRC
 451 for signature selection in the simulated data for each meta-analysis approach.

452 We conducted another set of simulation experiments for meta-analyzing studies with correlated
 453 samples. We first simulated independent samples as described above. We randomly picked
 454 two samples X_{i1} and X_{i2} to form cluster i . To introduce correlation between samples within

a cluster, we sample baseline composition X_i^* from the generalized Dirichlet distribution and merge this baseline composition into X_{i1} and X_{i2} and form the correlated samples $X_{i1}^* = (X_i^* + X_{i1})/2$ and $X_{i2}^* = (X_i^* + X_{i2})/2$ for cluster i . The case and control labels are randomly assigned to clusters. Differential abundance between cases and controls is introduced in the same manner as for independent samples.

Datasets of the five metagenomics studies for colorectal cancer (CRC). All raw sequencing data across studies were reprocessed using the same bioinformatics pipeline for taxonomic profiling³. The species-level RA count data and the variable for CRC status were obtained from https://github.com/zellerlab/crc_meta. The five studies involve subjects from five different countries. The sample sizes of the studies are 109, 127, 120, 114, and 104 (Table S1). We removed two samples with an outlier sequencing depth of less than 2,000. Our main analysis was on $K = 401$ species after removing species with a prevalence less than 20% (prevalence was defined as the percentage of non-zero counts in all samples across studies). Additionally, we analyzed the data with all $K = 849$ species and with $K = 267$ species under order *Clostridiales* to assess the stability of signature selection.

Datasets of the eight microbiome-metabolome studies. In all studies, the microbiome and metabolome were profiled from the same fecal samples. The datasets undergo curation and processing as part of an established effort⁶⁵. The genus-level RA count data and the variables for metabolites and disease status were obtained from <https://github.com/borenstein-lab/microbiome-metabolome-curated-data>. We used the eight adult cohorts from this resource with sample sizes of 96, 294, 240, 220, 444, 382, 287, and 164. There is notable heterogeneity among studies in study designs, microbiome sequencing methods, and metabolite measurements (Table S3). Our analysis focused on 450 metabolites with study-specific prevalence $\geq 10\%$ in at least two studies and 101 genera with study-specific prevalence $\geq 10\%$ in at least one study. We applied rank-based inverse normal transformation

481 to the metabolite variable. We performed Melody and MMUPHin to discover genus signatures
482 for each metabolite. In both methods, we analyze each metabolite variable as the covariate
483 of interest while adjusting the disease status as a confounder when generating summary
484 statistics within individual studies.

485 **Web resources.** R packages: *MMUPHin* (v1.14.0), *ALDEx2* (v1.32.0), *ANCOMBC* (v2.2.2),
486 *coin* (v1.4-2), *glmnet* (v4.1-8), *brglm2* (v 0.9.2), *abess* (v0.4.8), *mlr3* (v0.16.1)

487 **Data availability.** This study made use of 13 publicly available datasets. The five
488 metagenomics data for colorectal cancer were obtained from ([https://github.com/zellerlab/](https://github.com/zellerlab/crc_meta)
489 `crc_meta`) and the eight microbiome-metabolome data were obtained from ([https://github.](https://github.com/borenstein-lab/microbiome-metabolome-curated-data)
490 `com/borenstein-lab/microbiome-metabolome-curated-data`). All raw data and processed
491 data used for analysis are also available at <https://github.com/ZjpWei/Melody>.

492 **Code availability.** Melody is implemented in the R package miMeta publicly available
493 at <https://github.com/ZjpWei/miMeta>. The site also contains a complete set of manuals
494 and instructions. The scripts generating reported results are deposited at [https://github.](https://github.com/ZjpWei/Melody)
495 `com/ZjpWei/Melody`.

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Author Contributions

ZZT conceived the idea and oversaw the study. ZZT developed Melody with contributions from GC and ZW. ZW implemented the method, conducted numerical studies, and developed the miMeta R package. ZZT wrote the first version of the manuscript. GC and ZW also contributed to the writing.

Competing Interests

The authors declare no competing financial interests.

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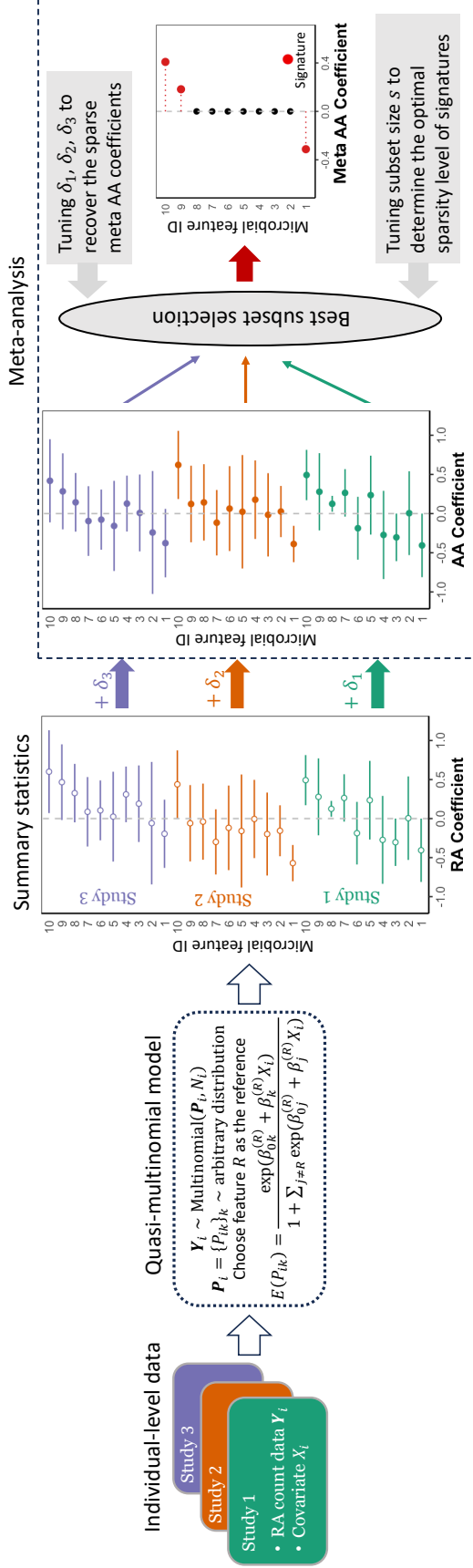


Fig. 1: Schematic overview of Melody. The Melody framework begins by generating summary statistics for each study using the quasi-multinomial model. In the model, the vector of true proportions $\mathbf{P}_i$ is allowed to follow an arbitrary distribution, and only its mean is modeled to link with the covariate of interest X_i . The $\beta_k^{(R)}$ is the relative abundance (RA) association coefficient of feature k given the reference feature R . The panel for summary statistics displays the estimates of the RA association coefficient (open dots) and the estimates plus or minus standard errors (horizontal lines). Next, the RA coefficient estimates in each study ℓ are shifted by δ_ℓ . These δ_ℓ values are jointly tuned with the sparsity level hyperparameter s for the best subset selection. In this illustration, the optimal value for δ_1 is close to zero, for δ_2 is positive, and for δ_3 is negative. As a result, the RA coefficients in Study 2 are shifted to the right, and those in Study 3 are shifted to the left to obtain the corresponding estimates of study-specific absolute abundance (AA) association coefficients (solid dots). These study-specific coefficients are then combined to estimate meta AA association coefficients. The optimal δ_ℓ values ensure that the RA coefficients are shifted appropriately to harmonize signals across studies and recover sparse meta AA associations.

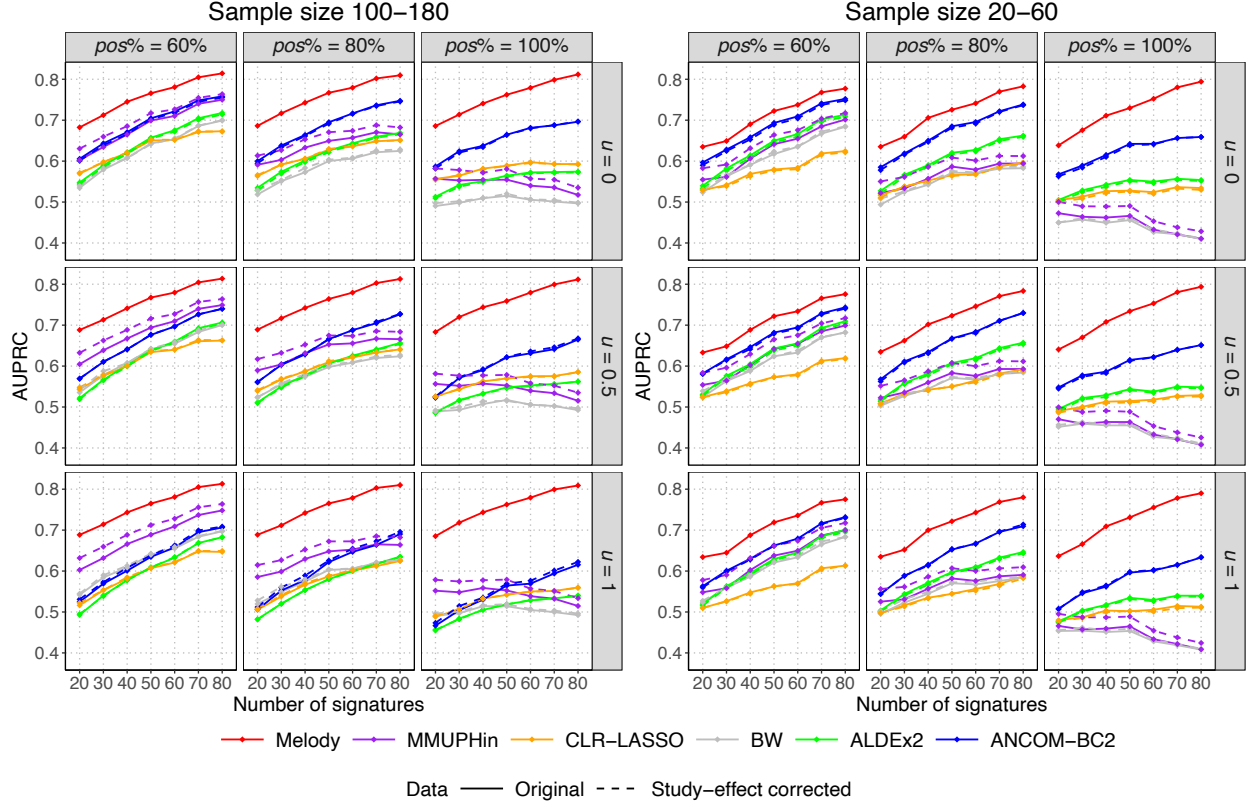


Fig. 2: Precision-recall evaluation of different meta-analysis signature selection approaches in simulated data with independent samples from five studies. Melody was applied to the original data of the five studies, and the other methods were applied to both the original data and study-effect corrected data using the MMUPHin batch-effect correction algorithm. The x -axis represents the number of signatures, and the y -axis shows the average area under the precision-recall curve (AUPRC) derived from 100 simulation replicates. The left and right sections show the results under different sample size settings (the sample size range for the five studies is 100-180 or 20-60). Within each section, the columns of the plots represent various degrees of sign imbalance among signatures ($pos\%$ denotes the percentage of signatures with positive effects); the rows of the plots represent various degrees of sequencing depth unevenness between the two groups of interest (u denotes the sequencing depth relative change ratio for the group with higher sequencing depth, and 0 indicates no unevenness).

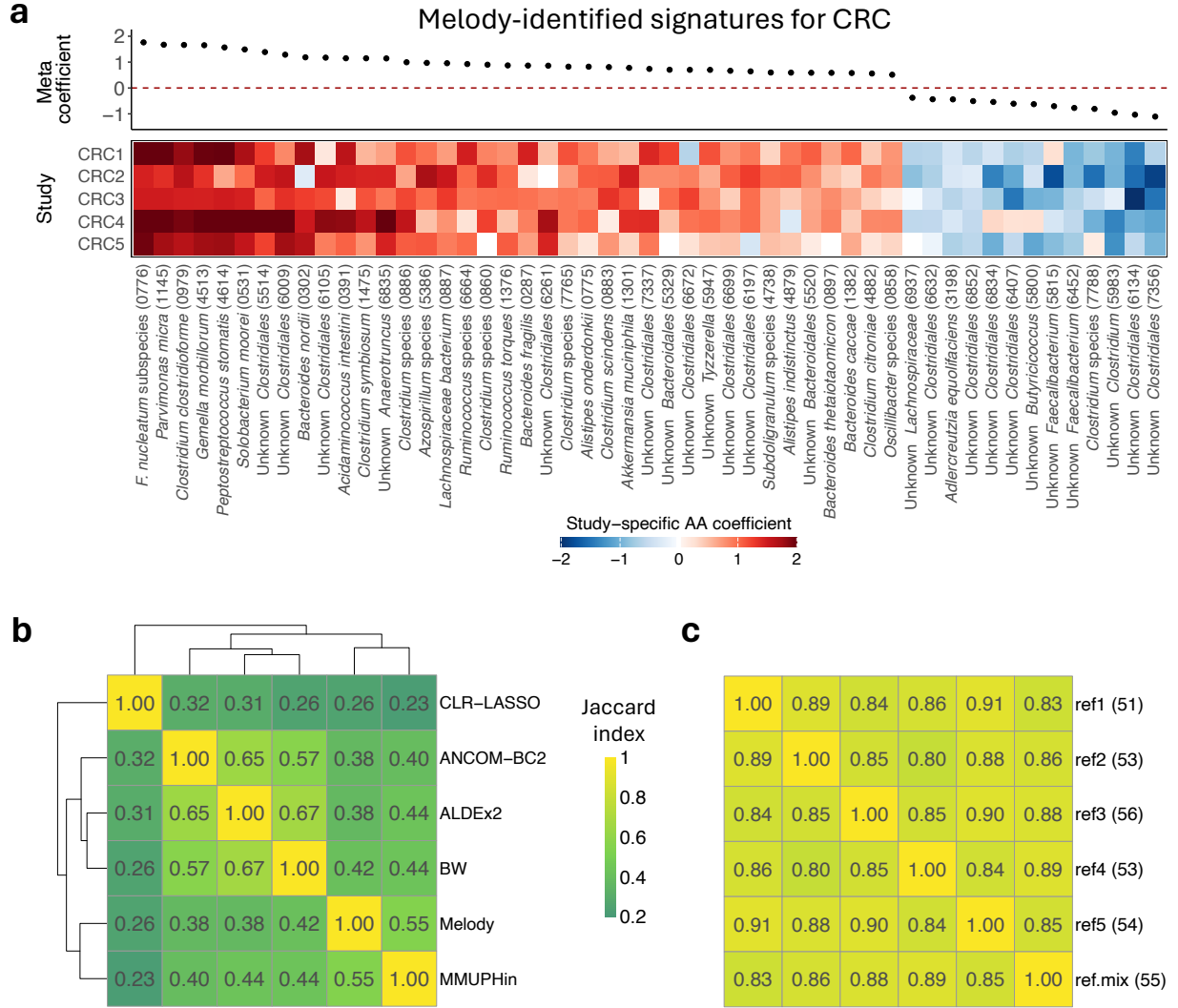


Fig. 3: Meta-analysis results from five metagenomics studies for identifying signatures for colorectal cancer (CRC). **a**, Estimates of the absolute-abundance (AA) association coefficient for the 51 signatures identified by Melody. The scatter plot displays the estimates of the meta coefficients, and the heatmap displays the estimates of the study-specific coefficients. **b**, Heatmap of the Jaccard index measuring the overlap of the top 51 selected species across different methods. The dendrogram is constructed using hierarchical clustering of the Jaccard index matrix. Results of all methods were generated on the original data. **c**, Heatmap of the Jaccard index measuring the overlap of the Melody-identified signatures across different reference configurations. The ref1 represents the scenario where we use species *Coprococcus catus* as the reference in all studies. The ref2-ref5 represent scenarios where we replace ref1 with a different reference and use the reference in all studies. The ref.mix represents the scenario where we use different references for different studies (ref1-5 for CRC1-5). The number in parentheses is the count of the identified signatures.

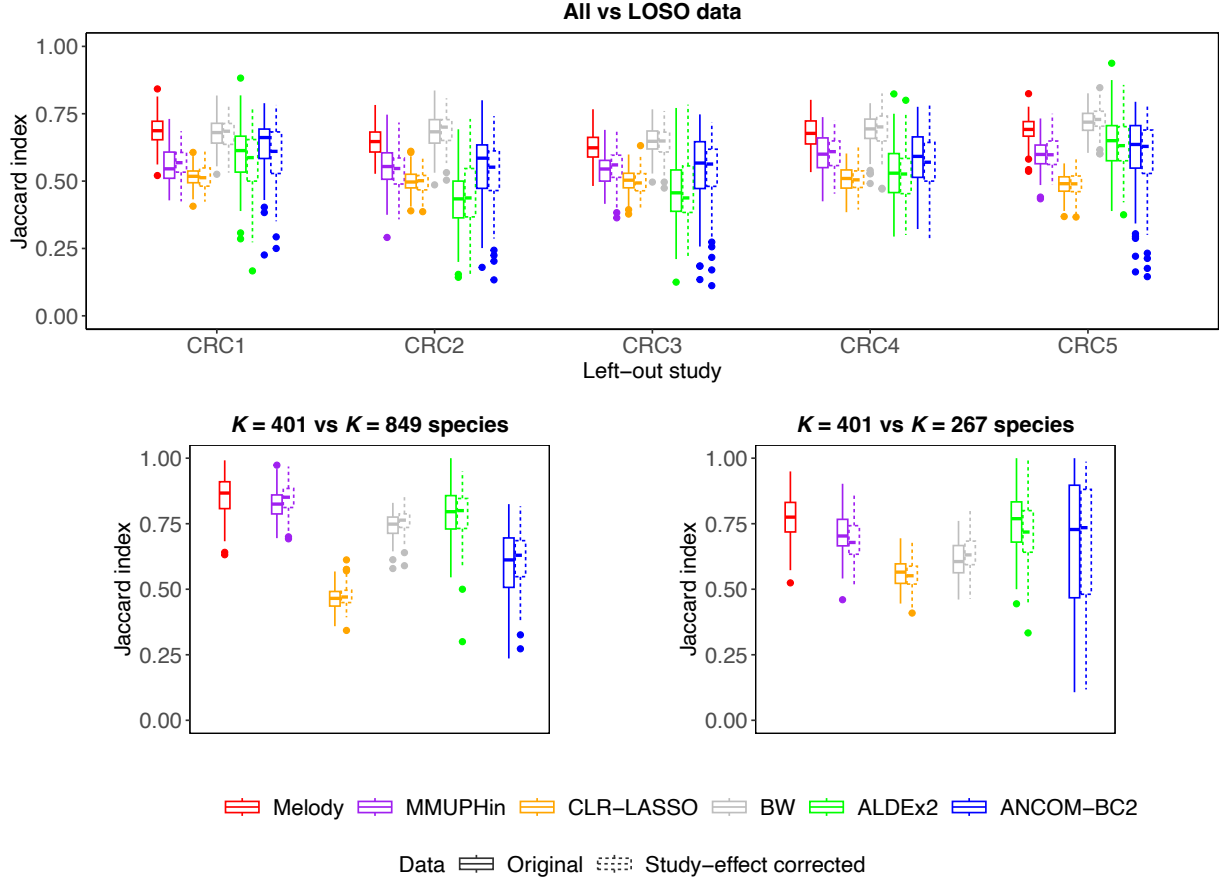


Fig. 4: Comparison of the stability of signature selection of different methods when performing two sets of meta-analyses based on metagenomics studies for colorectal cancer. Melody was applied to the original data, and the other methods were applied to both the original data and study-effect corrected data. The Jaccard index measures the overlap of the identified signatures from two sets of meta-analyses. Upper panel: Analysis 1 is based on all data from the five studies, and Analysis 2 is based on leave-one-study-out (LOSO) data. Lower left panel: Analysis 1 includes $K = 401$ species, and Analysis 2 includes all $K = 849$ species. Lower right panel: Analysis 1 includes $K = 401$ species, and Analysis 2 includes $K = 267$ species under order *Clostridiales*. Box plots were derived from 100 bootstrap samples of the complete data from the five studies. Each box indicates the median (center line), the first and third quartiles (box edges), whiskers extending to 1.5 times the interquartile range, and individual outliers (dots).

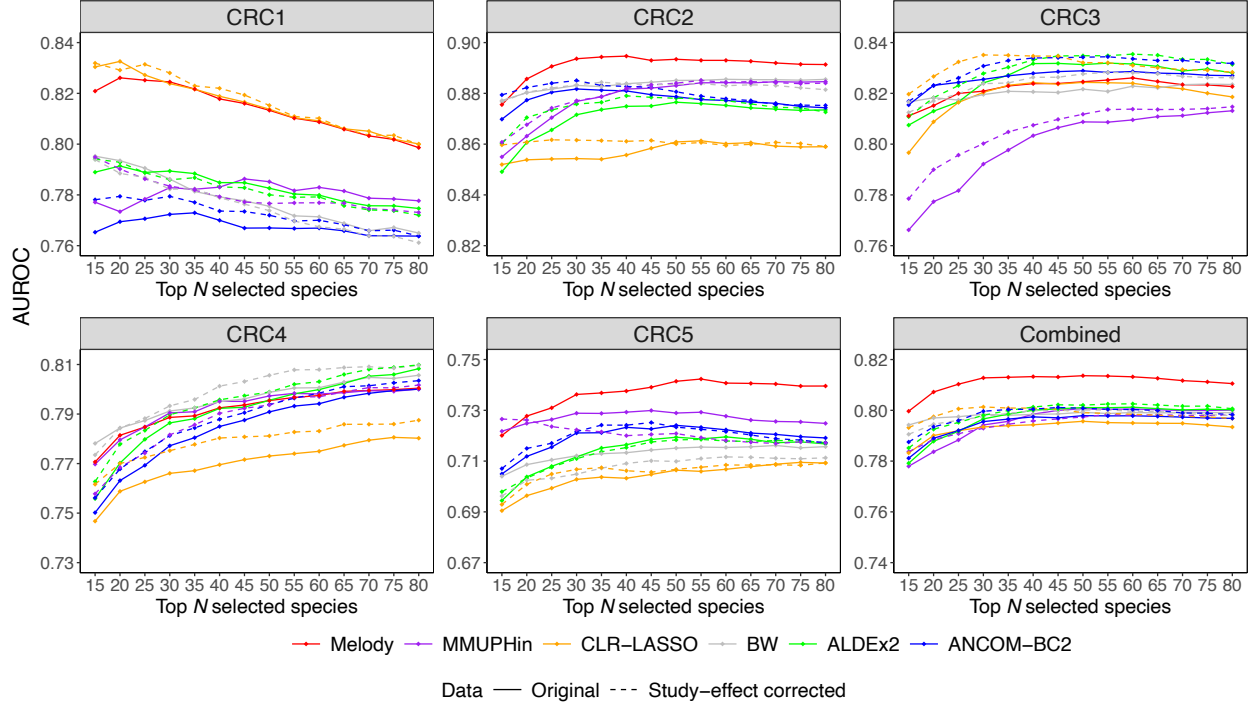


Fig. 5: Comparison of colorectal cancer (CRC) prediction using the top species identified by different meta-analysis methods. One study was held out each time for prediction evaluation, and samples from the remaining studies were randomly split into a feature selection set and a prediction model training set. Meta-analysis methods were applied to the feature selection set to identify the top N ($N = 15, \dots, 80$) CRC-associated species. Melody was applied to the original data, and the other methods were applied to both the original data and study-effect corrected data. A random forest model for CRC prediction was trained using the sample proportions of the top species in the training set, and the best model was evaluated on the hold-out study based on the area under the receiver operating characteristic (AUROC). The procedure was repeated 100 times with different random splits of the samples. In the first five panels, the average AUROC is provided in each panel, with the hold-out study ID listed at the top. The “Combined” panel displays the average AUROC across all hold-out studies.

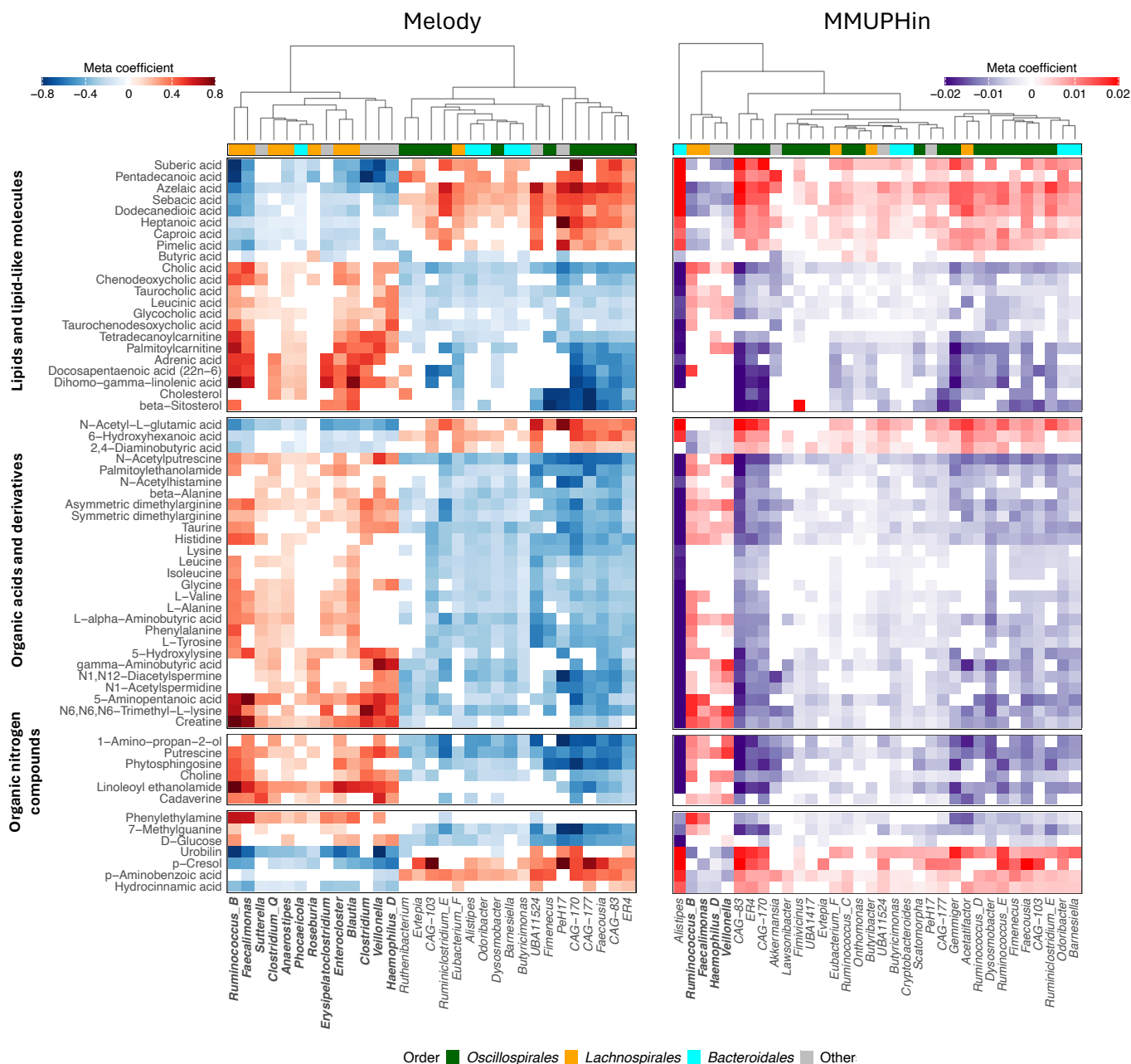


Fig. 6: Heatmaps of meta association coefficient estimates for the Melody and MMUPHin identified metabolite-genus pairs. The coefficient is shown as zero if the method does not select the metabolite-genus pair. The rows of both heatmaps are for the same set of top metabolites that are associated with at least 30 genera in Melody results. Most of the metabolites are grouped into three HMDB superclasses (names shown on the leftmost), while the remaining metabolites are placed at the bottom rows. The columns of the two heatmaps show genera associated with at least 70 metabolites, one from Melody results and the other from MMUPHin results. The names of the genera that have disproportionately positive associations with the top metabolites are in bold. Bacterial orders are summarized by the color code. Rows within each superclass and columns are clustered by Euclidean distance.